Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/15/2009
PI Name:	Bloomberg, Jacob J. Ph.D.		
Project Title:	Promoting Sensorimotor Response Generalizability: A Countermeasure to Mitigate Locomotor Dysfunction After Long-Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
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No. of Post Docs:	2	**No. of PhD Degrees:**	3
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: end date changed to 9/30/2008, from 9/01/2010, per PI and S. McCollum/JSC (10/08) In flight development phase (data collection has begun)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mulavara, Ajitkumar (University Space Research Association) Cohen, Helen (Baylor College of Medicine) Peters, Brian (Wyle Life Sciences) Miller, Chris (Wyle Life Sciences) Kozlovskaya, Inessa (Institute for Biomedical Problems)		
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Task Description:	Following their return to Earth, astronauts experience disturbances in their ability to walk and maintain postural stability due to neural adaptation to the microgravity conditions of space flight. These changes can impact mission objectives during planetary EVAs and may impair emergency vehicle egress capability. The goal of this project was to characterize the effects of space flight on the control of locomotion and develop and test an inflight balance and gait training program that would facilitate recovery of functional mobility after space flight. Due to logistic issues the inflight hardware that was to be used to provide the gait adaptability training on ISS treadmill exercise was not flown to the ISS. This report focuses on the pre/postflight component of this study that characterized the effects of long-duration space flight on the control of locomotion and dynamic visual acuity. We collected pre and postflight locomotion data on crewmembers from ISS Expeditions 5-12. Locomotor function was assessed before and after space flight using two tests of gait function. The Integrated Treadmill Locomotion Test (ITLT) characterized alterations in the integrated function of multiple sensorimotor sub-systems. Subjects walked on a motorized treadmill while we assessed changes in postural stability, head-trunk coordination, dynamic visual acuity and lower limb coordination strategies. The Functional Mobility Test (FMT) provided a corresponding assessment of the functional and operational changes in locomotor function by testing subjects' ability to negotiate an obstacle course placed over a medium-density foam floor. Results from the FMT indicate that adaptation to space flight led to a 62% increase in time to traverse the obstacle course on one day after return, and recovery of function took an average of 11 days after return. Results from the ITLT showed that subjects exhibited postflight changes in head-trunk coordination, toe clearance and gait cycle timing during treadmill locomotion Dynamic visual acuity was decreased postflight followed by an improvement in performance during the post flight recovery period. An analysis of the relationship between inflight treadmill usage characteristics and postflight functional mobility performance revealed that subjects who exercised on the iRED with greater body loading had enhanced postflight (R+1) functional mobility performance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data obtained in this study will aid in the design of a countermeasures system used to support exploration-class space missions. The operational version of this countermeasure will use a virtual reality (VR) system coupled with multi-direction treadmill along with haptic glove devices that will allow the user to walk in any direction and perform complex manual tasks in a varied and interesting virtual environment. This type of fusion interface, which incorporates both virtual and non-virtual devices across sensory modalities, produces multi-sensory, virtually augmented, synthetic environments. These synthetic environments can serve as pre-and inflight training tools providing sufficient sensorimotor challenge to crewmembers to maximize their motor response adaptability, facilitating the adaptive transition to partial or unit gravity. Within the context of locomotor functional training and rehabilitation, it is clear that an enriched and challenging environment provided by virtual reality devices provides the optimal conditions to explore variability and in that exploration increase adaptability of motor function. Indeed, the concept of employing virtual reality technology as a tool for rehabilitation has become a topic of great interest among a number of researchers. A collateral benefit of the application virtual reality technology, in this context, will be to make gait training programs for both space flight and rehabilitation more interesting, ultimately leading to increased adherence to prescribed training regimens. As people age on Earth, they sometimes experience instabilities in standing and walking. The development of unique walking and balance training procedures like the ones proposed in this study can be used to help prevent falling and injury in the elderly and clinical vestibular population. We have pursued this line of research in collaboration with Drs. Ronita Cromwell and Regina Buccello-Stout at the University of Texas Medical Branch in Galveston. This associated study compared a group of elders receiving the specialized treadmill adaptability training with a group that received the treadmill training only. Results from this study showed significantly improved gait function in elders that received the sensorimotor adaptability training (Buccello-Stout et al., 2008) . This study clearly demonstrated the general utility of the training method beyond the NASA applications.
	Conclusions The following conclusions address the risks and research gaps outlined in NASA's Human Research Program (HRP) Integrated Research Plan Addressing Research Gap SM2: "What is the time course of recovery of sensorimotor function after long-duration space flight?" 1) Conclusion: Adaptation to long-duration space flight led to a 62% increase in time to complete Functional Mobility Test obstacle course one day after landing (R+1). Recovery to pre-flight scores took an average of 11 days after landing. Implications: During the acute phase of adaptation to novel gravitational environments on exploration class missions sensorimotor disturbances will probably disrupt the ability to ambulate and perform functional tasks. These disturbances are likely to impair performance of complex tasks and could, therefore, affect astronaut safety and ultimate mission success. Also, current concepts for the crew return via the Constellation's Orion spacecraft after long-duration space flight include a water landing. This poses a unique risk to deconditioned and sensorimotor impaired crewmembers if an emergency egress is required soon after landing. Forward Work: Results obtained from the Functional Mobility Test served to motivate the design of the Functional Task Test (DSO 640), a pre/postflight battery of tests that will be used to assess performance of astronauts on functional tasks that are representative of critical mission tasks for lunar and Mars operations. 2) Conclusion: Recovery of functional mobility after long-duration space flight is composed of two distinct but interrelated motor learning processes: i) rapid on-line strategic change characterized by immediate onset after landing and ii) slower adaptive change requiring days to complete after landing.

Task Progress:	Implications: Postflight recovery in functional mobility involves both strategic and adaptive mechanisms and that there is dynamic interplay between these two processes. Adaptability training programs currently being developed as a countermeasure to facilitate adaptive transition to novel g environments should be designed to engage both strategic and adaptive systems to fully promote rapid and complete recovery of sensorimotor function. Forward Work: Current ground-based studies being conducted in our laboratory are focusing on developing adaptability training programs that include both gait and manual control training. In addition this research is investigating the retention capabilities of adaptability training allowing this type of training to be conducted preflight and therefore reducing the requirement for an inflight training component saving inflight crewtime and upmass hardware requirements. Addressing HRP Risk(s): Risk of Impaired Ability to Maintain Control of Vehicles and Other Complex Systems 3) Conclusion: Postflight (R+1) changes in gaze control produced decreases in dynamic visual acuity (DVA) during walking. For some subjects the decrement was greater than the mean acuity decrement seen in a population of vestibularly impaired patients collected using a similar protocol. The population mean showed a consistent improvement in DVA performance during the two-week postflight recovery period. Implications: While these data show changes in the ability of the crewmembers to see visual targets clearly while walking following space flight, these measures likely underestimate the decrements in visual performance that could be experienced during and immediately following landing. First, with the exception of one crewmember all of the DVA data were collected no earlier than 24 hrs after the landing. Given the rapid re-adaptation of gaze control we can infer that the decrement in visual acuity at landing was significantly higher. Also, walking is an active, well practiced, activity and therefore feed-forward control mechanisms facilitate the process of producing compensatory eye movements. The walking DVA test therefore underestimates the amount of decrement in visual performance that would be expected during passive, unpredictable motion that would occur in a landing vehicle. Forward Work: Post-flight neurological exams conducted by flight surgeons have rated gaze and ocular symptoms from mild to severe in over 50% of returning Shuttle astronauts. These observations are consistent with the decrements in visual acuity during locomotion described in this report. The Sensorimotor Discipline Implementation Team has recommended the addition of a “field” test of dynamic visual acuity that could be incorporated in the post-flight examinations. Incorporation of a post-flight dynamic visual performance assessment has been assigned a high priority as a research gap for the Sensorimotor Discipline. This research should focus on the development of a portable system that will provide passive body motion (i.e. seated subject on a motion base) while subjects perform tests of visual performance. Astronauts should be tested as soon as possible after landing to assess the early effects of gaze dysfunction related to sensorimotor adaptation. This will allow a better determination of the potential impacts of changes in visual acuity on landing performance. These data will inform the Constellation Program about expected changes in visual performance and will provide requirement for lander display panels. 4) Conclusion: Subjects exhibited postflight changes in head-trunk coordination, toe clearance, gait cycle timing and torso stability during treadmill locomotion. Implications: Changes in multiple sensorimotor control systems as a result of exposure to space flight leads to decrement in postflight functional mobility. Forward Work: The Functional Task Test (FTT, DSO 640) will allow better definition of the linkage between physiological change and functional performance including across several physiological systems (sensorimotor, cardiovascular, muscle performance). Addressing Gap SM1: "Relationship between the mode of in-flight exercise and post-flight sensorimotor performance?" 5) Conclusion: Relating the post-flight performance on the Functional Mobility Test with in-flight exercise produced one major observation: Subjects who performed squat exercises on the iRED with greater loads had enhanced post-flight (R+1) functional mobility performance. Implications: The association between loads lifted during squat exercises using the iRED and post-flight FMT performance could be due to the challenge offered to the postural muscles during these exercises on orbit with increased load. This provides enhanced postural variability at higher loads producing a static and dynamic balance challenge and perhaps serves to improve post-flight locomotor control. Current ground-based studies being conducted in our lab are investigating the role of increasing support surface instability during various resistance and locomotor exercises as a modality to enhance adaptability of sensorimotor function. Support surface variation during resistance and treadmill gait training is achieved by mounting a resistance exercise device and or a treadmill on a six-degree-of-freedom motion device. This allows motion of the support surface in multiple degrees of freedom and serves as a test bed to investigate novel resistance and treadmill training techniques for eventual application as part the nominal astronaut pre- or in-flight exercise training programs. Forward Work: It will be useful to examine the relationship between other in-flight exercise modalities (cycle ergometer, upper body resistive exercise) and post-flight functional mobility performance to determine if a relationship exists. The current in-flight exercise data set includes only average data from US crewmembers over the entire duration of the mission. The exercise prescriptions to US crewmembers are graded over the duration of mission with the intensity (especially loading) and duration increased towards the end of the mission. Hence, the above analysis may be over conservative in its estimates and should be extended with data over separate in-flight periods to confirm the relationships observed in the present study.
Bibliography Type:	Description: (Last Updated: 11/14/2025)

Abstracts for Journals and Proceedings	Miller C, Peters B, Brady R, Mulavara A, Richards J, Hayat M, Bloomberg J. "Effects of long-duration space flight on toe clearance during treadmill walking." Gait II session, 2008 North American Congress on Biomechanics, Ann Arbor, MI, August, 2008. 2008 North American Congress on Biomechanics, Ann Arbor, MI, August, 2008. , Aug-2008
Abstracts for Journals and Proceedings	Bloomberg JJ, Mulavara AP, Peters BT, Miller CA, Richards JT, Brady R, Cohen HS, Kozlovskaya IB. "Motor learning processes mediating recovery of functional mobility after space flight." Voluntary Movement: Plasticity II session, Annual Meeting of the Society for Neuroscience, Washington, DC, November 15-19, 2008. Annual Meeting of the Society for Neuroscience, Washington, DC, November 15-19, 2008. , Nov-2008
Abstracts for Journals and Proceedings	Bloomberg JJ, Peters BT, Mulavara AP, Ruttley TM, Miller CA, Brady R, Cohen HS. "Development of sensorimotor adaptability training programs." NASA Human Research Program Investigators' Workshop, Houston, TX, February 2008. NASA Human Research Program Investigators' Workshop, Houston, TX, February 2008. , Feb-2008
Abstracts for Journals and Proceedings	Mulavara AP, Cohen HS, Peters BT, Miller CA, Brady R, Bloomberg JJ. "Locomotor Dysfunction after Spaceflight: Characterization and Countermeasure Development." Motor Control session, Houston Society for Engineering in Medicine and Biology 23rd Annual Houston Conference on Biomedical Engineering Research, Houston, TX, 9-10 February 2007. Houston Society for Engineering in Medicine and Biology 23rd Annual Houston Conference on Biomedical Engineering Research, Houston, TX, 9-10 February 2007. , Feb-2007
Abstracts for Journals and Proceedings	Bloomberg JJ, Mulavara AP, Peters BT, Cohen HS, Miller CA, Brady R, Warren LE, Ruttley TM, Kozlovskaya IB. "Development of training programs to optimize planetary ambulation." NASA Human Research Program Investigators' Workshop, Houston, Texas, February 12–14, 2007. NASA Human Research Program Investigators' Workshop, Houston, Texas, February 12–14, 2007. , Feb-2007
Abstracts for Journals and Proceedings	Peters BT, Van Emmerik R, Brady R, Bloomberg JJ. "Visual acuity during treadmill walking." Visual Functions session, the18th Meeting of the International Society for Posture and Gait Research, Burlington, VT, July 15-18, 2007. 18th Meeting of the International Society for Posture and Gait Research, Burlington, VT, July 15-18, 2007. , Jul-2007
Abstracts for Journals and Proceedings	Miller C, Peters B, Brady R, Mulavara A, Warren L, Feiveson A, Bloomberg J. "Comparison of two alternate methods for tracking toe trajectory." Locomotion session, 31st Annual Meeting of the American Society of Biomechanics, Stanford University, Palo Alto, CA, August 2007. 31st Annual Meeting of the American Society of Biomechanics, Stanford University, Palo Alto, CA, August 2007. , Aug-2007
Articles in Peer-reviewed Journals	Miller CA, Feiveson AH, Bloomberg JJ. "Effects of speed and visual-target distance on toe trajectory during the swing phase of treadmill walking." Journal of Applied Biomechanics, 2009 Feb;25(1):32-42. PubMed PMID: 19299828 , Feb-2009